A Bayesian Neural Network Surrogate for Coupled Neutronics–Thermal–Structural Uncertainty Analysis: Application to a Heat-Pipe-Cooled Reactor

面向耦合中子–热–力不确定性分析的贝叶斯神经网络代理模型——以热管冷却反应堆为例

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摘要 / Abstract

We develop a Bayesian neural network (BNN) surrogate for coupled neutronics–thermal–structural analysis in a heat-pipe-cooled reactor (HPR). The BNN is trained on coupled OpenMC–FEniCS simulations and maps eight uncertain material parameters to a posterior predictive distribution that is propagated consistently through forward uncertainty analysis, global sensitivity attribution and observation-conditioned posterior calibration. On a held-out test set the surrogate attains stress $R^2 = 0.910$ with conservatively covered 90% predictive intervals adequate for distribution-level downstream analysis. Forward Monte Carlo propagation shows that multiphysics coupling reshapes the stress distribution, reducing its standard deviation by ≈ 28% relative to the decoupled prediction. Sobol attribution identifies largely separated dominant pathways: $E_{\mathrm{intercept}}$ for stress ($S_1 \approx 0.56$) and α_{base} for k_{eff} ($S_1 \approx 0.74$). MCMC calibration contracts the inferred parameter distribution toward observation-compatible regions (90% coverage 0.889 across 18 benchmark cases). A single posterior-predictive draw costs ≈ $1.76 \times 10^5\times$ less than a

coupled high-fidelity case (2266 s on AMD EPYC 9654), making the full probabilistic workflow tractable at the surrogate level.

本文面向中子–热–力双向耦合的热管冷却反应堆（HPR），提出一套贝叶斯神经网络（BNN）概率代理建模框架。BNN 在耦合 OpenMC–FEniCS 仿真数据上训练，将 8 个材料不确定输入映射为后验预测分布，并将该分布一致地用于前向不确定性分析、全局敏感性归因与观测条件下的后验标定。在留出测试集上，代理模型应力 $R^2 = 0.910$，90% 后验预测区间呈保守覆盖，精度水平足以支撑分布层面的下游分析。前向蒙特卡洛传播表明，多物理耦合重塑应力分布，使其标准差相对解耦预测降低约 28%。Sobol 归因显示应力与 k_{eff} 的主导路径基本独立（应力：$E_{\mathrm{intercept}}$，$S_1 \approx 0.56$；k_{eff}：α_{base}，$S_1 \approx 0.74$）。MCMC 标定将推断参数分布收敛至与观测相容的区域（18 例基准案例 90% 覆盖率 0.889）。单次后验预测抽样耗时较耦合高保真求解器缩短约 1.76×10^5 倍（HF 单案例 2266 s，AMD EPYC 9654 实测基线），使完整概率工作流在代理层级上可行。

1. 引言 / Introduction

Heat-pipe-cooled reactors (HPRs) are compact, monolithic-core systems in which neutronics, heat transfer and structural deformation are coupled through two-way feedback rather than arranged in a one-way sequence [1]. Temperature fields drive thermal expansion and stress, and the resulting geometric changes feed back into neutron leakage and power distribution. Under such closed feedback, uncertainty in material properties propagates across coupled neutronic, thermal and structural channels rather than within a single physics: the same material parameter can influence multiple outputs simultaneously, and the coupled feedback may either amplify or attenuate the resulting response variability. A coupled probabilistic treatment, rather than decoupled or single-physics evaluation, is therefore required.

热管冷却反应堆（HPR）是将中子学、传热与结构变形通过双向反馈紧密耦合的紧凑型基体式堆芯系统，而非单向串联过程 [1]。温度场驱动基体热膨胀与应力演化，由此产生的几何变化又反馈至中子泄漏与功率分布。在这种闭合反馈下，材料属性不确定性并非局限

于单一物理，而是在耦合的中子、热工与结构通道间同时传播：同一材料参数可同时作用于多个输出量，而耦合反馈既可能放大、也可能衰减最终响应变异。因此必须采用耦合概率处理，而不能依赖解耦或单场评估。

High-fidelity multiphysics analysis of compact heat-pipe microreactors has become an active international direction. MOOSE-based toolchains that couple neutronics, thermo-mechanics and heat-pipe physics have been demonstrated on both KRUSTY-class and larger heat-pipe reactor designs [2,3]. Parallel studies based on commercial thermal–structural solvers repeatedly identify stress concentration and structural integrity as recurring safety concerns in monolithic-core configurations [4,5]. Beyond the HPR literature, nuclear computational science is now developing stochastic analysis, surrogate generation and probabilistic workflow capabilities around large-scale simulators, both within the MOOSE ecosystem [6,7] and through emerging digital-twin-oriented pipelines [8]. These advances nonetheless stop short of a unified probabilistic computational layer that, within a single coupled framework, simultaneously supports fast surrogate prediction, forward uncertainty propagation, global sensitivity attribution and observation-conditioned posterior updating.

紧凑型热管微堆的高保真多物理分析已成为国际上的活跃方向。基于 MOOSE 的工具链已在 KRUSTY 类与更大功率热管堆设计上实现了中子学、热–力学与热管物理的耦合 [2,3]。基于商用热–力求解器的平行研究反复指出，应力集中与结构完整性是基体式堆芯构型中的持续性安全关注点 [4,5]。除 HPR 专门研究外，核工程计算领域也正在围绕大规模仿真器发展随机分析、代理建模与概率工作流能力，其中既包括 MOOSE 生态中的相关模块 [6,7]，也包括面向数字孪生的初步管线 [8]。然而这些进展仍未形成一个统一的概率计算层，尚不能在同一耦合框架内同时支撑快速代理预测、前向不确定性传播、全局敏感性归因与观测条件下的后验更新。

Direct evaluation based on repeated high-fidelity simulation is computationally prohibitive: each coupled OpenMC–FEniCS solve requires on the order of 10^3 s per sample, so forward Monte Carlo over an 8-dimensional material-parameter space, and Bayesian calibration requiring comparable likelihood counts, are both infeasible at the high-fidelity level. Standard surrogates close only part of this gap. Gaussian-process and polynomial-chaos methods become increasingly

constrained in strongly nonlinear, moderately high-dimensional coupled settings; physics-informed neural networks are difficult to formulate when the governing structure is expressed through iterative solver exchange rather than a single closed residual; and purely deterministic data-driven networks scale but, lacking a coherent predictive distribution, do not provide a single object that can be reused across forward propagation, sensitivity attribution and observation-conditioned updating. Bayesian neural networks (BNNs) place a distribution over the network weights, so that the resulting posterior predictive distribution can be evaluated densely on the surrogate side and can serve as the common probabilistic object on which downstream forward UQ, sensitivity and posterior analyses operate.

直接基于高保真重复求解的评估在计算上不可行：每次 OpenMC–FEniCS 耦合求解约需 10^3 秒，因而在 8 维材料参数空间上进行前向蒙特卡洛传播、以及需要相近量级似然评估的贝叶斯标定，均无法在高保真层面完成。常规代理方法仅能部分弥合这一差距：高斯过程与多项式混沌方法在强非线性、中高维耦合设置下受维度与光滑性假设限制；物理信息神经网络难以为以迭代求解器交换而非单一闭合残差表达的控制结构直接构造损失；纯确定性数据驱动神经网络虽具扩展性，但缺乏连贯的预测分布，无法提供一个可在前向传播、敏感性归因与观测条件后验更新之间复用的统一对象。贝叶斯神经网络（BNN）通过对网络权重引入分布，使后验预测分布可在代理侧被密集采样，并作为下游前向 UQ、敏感性与后验分析所共用的概率对象。

In this work we develop a Bayesian neural network probabilistic surrogate for a coupled neutronics–thermal–structural system and demonstrate it on a heat-pipe-cooled reactor. The BNN is trained on coupled OpenMC–FEniCS data and produces, for multiple physics outputs, a posterior predictive distribution that is then propagated consistently through forward Monte Carlo analysis, global sensitivity attribution with bootstrap confidence intervals, and MCMC posterior calibration against noisy observations. The main contributions are threefold. First, we introduce a BNN probabilistic surrogate for the coupled neutronics–thermal–structural response that delivers calibrated posterior predictive distributions for the coupled outputs at evaluation cost amenable to dense sampling. Second, we organise forward uncertainty propagation, Sobol sensitivity attribution and observation-conditioned MCMC calibration on top of this same predictive distribution, so the three analyses share a single probabilistic object rather than being driven by

separately tuned point models. Third, we demonstrate that this BNN posterior predictive layer connects expensive coupled multiphysics simulation to a tractable end-to-end probabilistic workflow in coupled HPR analysis.

本文面向耦合中子–热–力系统，构建了一种贝叶斯神经网络概率代理模型，并以热管冷却反应堆作为工程验证对象。BNN 在耦合 OpenMC–FEniCS 数据上训练，对多个物理输出给出后验预测分布，并将该分布一致地用于前向蒙特卡洛分析、带 Bootstrap 置信区间的 Sobol 全局敏感性归因与含噪观测下的 MCMC 后验标定。主要贡献有三：其一，提出面向耦合中子–热–力响应的 BNN 概率代理模型，对耦合输出给出经校准的后验预测分布，且评估代价足以支持密集采样；其二，在同一后验预测分布之上组织前向不确定性传播、Sobol 敏感性归因与观测条件后验标定，使三类分析共用同一概率对象，而非分别依赖独立调参的点估计模型；其三，论证该 BNN 后验预测层将昂贵的耦合多物理仿真衔接至耦合 HPR 分析中可行的端到端概率工作流。

2. 结果 / Results

2.1 代理精度 / Surrogate Accuracy

The BNN surrogate places a distribution over network weights and yields a posterior predictive distribution for each output; architecture, prior, and inference scheme are detailed in Methods. The surrogate was evaluated on the held-out test set (n = 435). For the primary safety metric, coupled steady-state maximum global stress, the BNN achieved $R^2 = 0.910$, RMSE = 8.93 MPa, and MAE = 6.98 MPa. The 90% posterior-predictive interval coverage $PICP_{90} = 0.998$ with mean interval width $MPIW_{90} = 64.3$ MPa indicates conservative over-coverage: the intervals are broader than the nominal 90% level, adequate for the distribution-level downstream analyses of Sections 2.2–2.4 but not sharply calibrated (Table 2, Fig. 2A). For k_{eff}, the BNN achieved $R^2 = 0.825$, $PICP_{90} = 1.000$, and $MPIW_{90} = 2.90 \times 10^{-3}$ (≈ 290 pcm), exhibiting the same conservative over-coverage pattern (Table 2).

本文采用的 BNN 代理模型在网络权重上引入分布，对每个输出量给出后验预测分布，具体架构、先验与推断方案见方法节。在留出测试集（n = 435）上，耦合稳态最大全局应力的预测精度为 $R^2 = 0.910$，RMSE = 8.93 MPa，MAE = 6.98 MPa； 90% 后验预测区间覆盖率 $PICP_{90} = 0.998$，平均区间宽度 $MPIW_{90} = 64.3$ MPa，表明区间呈保守偏覆盖（over-coverage）：相对标称 90% 水平偏宽，足以支撑 §2.2–§2.4 的分布级下游分析，但并非紧贴名义水平的精准校准（表 2，图 2A）。k_{eff} 预测精度为 $R^2 = 0.825$，$PICP_{90} = 1.000$，$MPIW_{90} = 2.90 \times 10^{-3}$（约 290 pcm），呈现同样的保守偏覆盖模式（表 2）。

An ablation that removes the monotonicity and inequality priors (an ELBO-only BNN trained on the same data) attains a comparable stress R^2 but yields noticeably wider 90% predictive intervals and weaker rank-monotonic behaviour with respect to the dominant parameters; the ablation is reported in the Appendix and is not used in any downstream UQ, sensitivity, or posterior analysis. Two properties of the constraint-regularised BNN therefore motivate its selection as the primary surrogate. First, the constraint terms tighten the predictive intervals relative to the unconstrained ablation while keeping coverage at the nominal level — i.e., they sharpen rather than miscalibrate the predictive uncertainty. Second, the BNN exposes a single posterior predictive distribution that is then propagated, without retraining or recalibration, through forward UQ, Sobol attribution and MCMC posterior calibration — making the three downstream analyses share a common probabilistic object rather than three independently tuned point models.

附录另报告一项消融对照：去除单调性与不等式先验、仅保留 ELBO 的 BNN 在同一训练集上达到可比的应力 R^2，但 90% 预测区间显著更宽、对主导参数的秩单调性更弱；该消融不参与下游 UQ、敏感性或后验分析。约束正则化 BNN 之所以被选为主代理，基于其两点性质：其一，约束项相对无约束消融收紧了预测区间，且覆盖率仍维持在名义水平——即约束项是在锐化预测不确定性而非引入失校；其二，BNN 暴露出一个单一的后验预测分布，可在不重训、不重校准的前提下被前向 UQ、Sobol 归因与 MCMC 后验标定共同复用——使下游三类分析以同一概率对象为基础，而非依赖三个独立调参的点估计模型。

2.2 前向不确定性传播 / Forward Uncertainty Propagation

Forward propagation of 20,000 Latin hypercube samples through the BNN posterior predictive mean function (y = E_w[μ_w(θ)], isolating parametric uncertainty from the BNN's epistemic+aleatoric predictive spread; see Methods) reveals how multi-physics coupling reshapes the output uncertainty landscape (Table 4, Fig. 3).

对 20,000 个拉丁超立方样本通过 BNN 后验预测均值函数（y = E_w[μ_w(θ)]，隔离参数不确定性与 BNN 自身认识论+偶然预测展宽；详见方法节）的前向传播，揭示了多物理耦合如何重塑输出不确定性格局（表 4，图 3）。

For coupled steady-state stress, the prior-predictive distribution has a mean of 162.5 MPa and standard deviation of 35.5 MPa (5th, 50th, 95th percentiles: 107.4, 160.5, 224.1 MPa). The corresponding decoupled (single-pass) prediction has a mean of approximately 202.4 MPa and standard deviation of approximately 49.2 MPa (derived from the test-set iter1/iter2 variance ratio var(σ_coupled)/var(σ_decoupled) = 0.52 and the observed mean shift Δ = −39.9 MPa between passes). The coupling feedback therefore reduces the stress standard deviation by ~28% (from ~49.2 to 35.5 MPa), suggesting that multi-physics coupling attenuates a substantial fraction of the material-parameter-driven stress variability that a decoupled analysis would predict.

耦合稳态应力的先验预测分布均值为 162.5 MPa，标准差 35.5 MPa（5、50、 95 分位数分别为 107.4、160.5、224.1 MPa）。对应的解耦（单轮）预测均值约为 202.4 MPa，标准差约为 49.2 MPa（由测试集 iter1/iter2 方差比 var(σ_coupled)/var(σ_decoupled) = 0.52 与观测到的平均差 Δ = −39.9 MPa 推得）。耦合反馈使应力标准差降低约 28%（从约 49.2 降至 35.5 MPa），表明多物理耦合衰减了解耦分析所预测的材料参数驱动应力变异中相当一部分。

For coupled k_{eff}, the standard deviation is 1.07×10^{-3} (approximately 107 pcm), with a mean of 1.1036; the corresponding decoupled (single-pass) k_{eff} is effectively constant across the LHS design (standard deviation ~5×10^{-14}), because the eight sampled material parameters do not enter the neutron cross-section tables at the reference geometry. The coupled k_{eff} spread is therefore attributable to thermo-mechanical feedback through thermal-expansion modification of

core geometry; no decoupled-to-coupled compression ratio is reported, since the decoupled value carries no meaningful parametric uncertainty.

耦合 k_{eff} 标准差为 1.07×10^{-3}（约 107 pcm），均值 1.1036；对应的解耦（单轮）k_{eff} 在 LHS 设计下实质上为常数（标准差约 5×10^{-14}），因为 8 个采样材料参数在参考几何条件下不进入中子截面表。耦合 k_{eff} 的展宽可归因于热膨胀改变堆芯几何这一热力反馈通道；由于解耦值不携带有意义的参数不确定性，不引用 k_{eff} 的解耦–耦合压缩比。

Taken together, the forward-UQ results show that multiphysics coupling does not merely shift the stress distribution but actively reshapes it: the standard deviation contracts substantially, while the median moves to lower values (162.5 MPa coupled vs. 202.4 MPa decoupled). The coupled prior-predictive stress distribution itself — characterised above by its mean, standard deviation and 5/50/95 percentiles — is the object on which the global sensitivity and posterior-calibration analyses of Sections 2.3 and 2.4 operate.

综合来看，前向 UQ 结果表明多物理耦合并非简单平移应力分布，而是对其进行实质性重塑：标准差显著收缩，且中位数同时向低值移动（耦合 162.5 MPa vs. 解耦 202.4 MPa）。上文以均值、标准差与 5/50/95 分位数所表征的耦合先验预测应力分布本身，即是 §2.3 全局敏感性分析与 §2.4 后验标定所共同作用的对象。

2.3 全局敏感性分析 / Global Sensitivity Analysis

Sobol variance-based sensitivity analysis (Saltelli sampling, Jansen estimator, N = 4096, 50 independent replications; see Methods) identified distinct dominant pathways for the two primary outputs (Table 5, Fig. 4).

基于方差的 Sobol 敏感性分析（Saltelli 采样、Jansen 估计量、N = 4096、 50 次独立重复；详见方法节）识别出两个主要输出的不同主导路径（表 5，图 4）。

For coupled steady-state stress, $E_{intercept}$ is the dominant first-order factor (S_1 = 0.558, 90% CI: [0.551, 0.565]), accounting for approximately 59% of the first-order variance. α_{base} ranks second (S_1 = 0.148, CI: [0.139, 0.157]), followed by SS316_k_ref (S_1 = 0.076, CI: [0.065, 0.087]) and ν (S_1 = 0.067, CI: [0.058, 0.077]). The remaining four inputs have S_1 confidence intervals approaching or including zero; factors whose CI crosses zero (SS316_alpha for stress in three of

four BNN variants) are not treated as stable contributors. The total-order index for $E_{intercept}$ (S_T = 0.597) exceeds S_1 by only 0.039, indicating that $E_{intercept}$ affects stress primarily through an independent channel with limited higher-order interaction.

耦合稳态应力：$E_{intercept}$ 是最主导的一阶因子（S_1 = 0.558，90% CI： [0.551, 0.565]），占一阶方差约 59%。α_{base} 居第二（S_1 = 0.148， CI：[0.139, 0.157]），SS316_k_ref 第三（S_1 = 0.076，CI：[0.065, 0.087]），ν 第四（S_1 = 0.067，CI：[0.058, 0.077]）。其余 4 个输入的 S_1 置信区间接近或包含零；CI 跨零的因子（四个 BNN 变体中三个的 SS316_alpha 对应力）不被视为稳定贡献。$E_{intercept}$ 的总阶指数（S_T = 0.597）仅比 S_1 高 0.039，表明其主要通过独立通道影响应力，高阶交互作用有限。

For coupled k_{eff}, α_{base} is the single dominant factor (S_1 = 0.742, CI: [0.736, 0.747]), followed by α_{slope} as a secondary contributor ($S_1 \approx 0.164$, CI: [0.150, 0.178]) and ν as a third stable contributor ($S_1 \approx 0.055$, CI: [0.042, 0.068]). The remaining five inputs have small S_1 values with confidence intervals including zero. The dominance of α_{base} reflects that thermal expansion governs core geometry — and thereby neutron leakage probability — which is the principal physical mechanism determining k_{eff} variation.

耦合 k_{eff}：α_{base} 是主控因子（S_1 = 0.742，CI：[0.736, 0.747]），α_{slope} 为次级贡献者（$S_1 \approx$ 0.164，CI：[0.150, 0.178]），ν 为第三位稳定贡献者（$S_1 \approx 0.055$，CI：[0.042, 0.068]）。其余 5 个参数的 S_1 值较小且置信区间包含零。α_{base} 的主导性反映热膨胀控制堆芯几何形状——进而控制中子泄漏概率——这是决定耦合稳态 k_{eff} 变异的主要物理机制。

The separation of dominant factors — $E_{intercept}$ for stress, α_{base} for k_{eff}, with limited cross-interaction (S_T − S_1 small for both) — indicates that the two primary outputs are governed by largely non-overlapping physical pathways at the dominant-factor level.

主控因子的分离——$E_{intercept}$ 主导应力、α_{base} 主导 k_{eff}，交叉作用有限（两者 S_T − S_1 均较小）——表明两个主要输出在主控因子层面由基本不重叠的物理路径支配。

2.4 后验标定 / Posterior Calibration

MCMC posterior calibration (Metropolis-Hastings, 8000 iterations, burn-in 2000, thinning 5, proposal scale 0.40; see Methods) was performed on 18 benchmark cases drawn from the held-out test split, spanning a low / intermediate / high stress band (six cases per band) so that the calibration exercise covers the operating range rather than a single nominal point. Observations are constructed by perturbing the true high-fidelity outputs of each test case with 2% Gaussian noise (likelihood model in Methods). Acceptance rates ranged from 0.65 to 0.67 (mean 0.66); trace-plot inspection supported stationarity after burn-in (Fig. A1). Multi-chain Gelman–Rubin $\hat{R}$ diagnostics are noted as pending: the per-case benchmark CSV currently has an empty $\hat{R}$ column that will be populated from multi-chain trace logs in the revision.

MCMC 后验标定（Metropolis-Hastings，8000 步迭代，烧入 2000，抽稀 5，提议尺度 0.40；详见方法节）在 18 个基准案例上执行，案例来自留出测试集，按低 / 中 / 高应力带各 6 例分层抽样，使标定覆盖运行范围而非单一标称点。观测量由各测试案例的真实高保真输出叠加 2% 高斯噪声构造（似然模型见方法节）。接受率 0.65–0.67（均值 0.66）；trace plot 确认烧入后平稳（图 A1）。多链 Gelman–Rubin $\hat{R}$ 诊断当前列为待补：per-case benchmark CSV 中 $\hat{R}$ 列为空，修订版将由多链 trace 日志补齐。

Across the 18 benchmark cases, the posterior 90%-CI coverage for the four calibrated parameters ($E_{\text{intercept}}$, α_{base}, α_{slope}, k_{ref}) was 0.889, indicating that the inferred parameter intervals encompass the true parameter values at approximately the nominal 90% rate. The marginal posteriors exhibit non-Gaussian features consistent with the multiplicative stress structure $\sigma \propto E \cdot \alpha \cdot \Delta T$: in higher-stress cases the $E_{\text{intercept}}$ marginal shifts toward larger values (consistent with the Sobol attribution in Section 2.3), and the joint posterior of $E_{\text{intercept}}$ and α_{base} develops a compensation ridge reflecting partial structural non-identifiability of the product form.

18 个基准案例上，4 个被标定参数（$E_{\text{intercept}}$、α_{base}、α_{slope}、k_{ref}）的后验 90%-CI 覆盖率为 0.889，表明所推断的参数区间以接近标称 90% 的比例覆盖真实参数值。边际后验呈现与乘积应力结构 $\sigma \propto E \cdot \alpha \cdot \Delta T$ 一致的非高斯特征：在较高应力案例中，$E_{\text{intercept}}$ 边际向更大

值偏移（与 §2.3 Sobol 归因一致），$E_{\text{intercept}}$ 与 α_{base} 的联合后验出现反映乘积形式部分结构不可识别性的补偿脊。

To verify that the inferred parameter distribution corresponds to a physically consistent state of the surrogate, we evaluated the BNN posterior-predictive stress at the posterior-mean parameters across the benchmark cases. The posterior-mean stress tracks the true stress with MAE = 6.6 MPa, RMSE = 7.2 MPa and mean bias +2.5 MPa over a true-stress range of 133–194 MPa, indicating that the MCMC posterior contracts the inferred parameter distribution toward observation-compatible regions of parameter space rather than wandering into surrogate-incompatible regions.

为核验所推断的参数分布对应于代理意义下物理一致的状态，我们在基准案例上将 BNN 后验预测应力评估于后验均值参数处。在真实应力范围 133–194 MPa 下，后验均值应力与真实应力吻合：MAE = 6.6 MPa，RMSE = 7.2 MPa，平均偏差 +2.5 MPa，表明 MCMC 后验将所推断的参数分布收敛至与观测相容的参数空间区域，而非漂移到与代理不相容的区域。

An independent high-fidelity consistency check — re-evaluating the 18 benchmark cases of Section 2.4 by the coupled OpenMC–FEniCS solver at their BNN posterior-mean parameter vectors — is in preparation and will be reported as supplementary material; it is not part of the present evidence base.

§2.4 全部 18 个基准案例将在 BNN 后验均值参数向量处由耦合 OpenMC–FEniCS 求解器重新评估的独立高保真一致性抽查正在准备中，将作为补充材料报告；该抽查不构成本文当前证据基础的组成部分。

3. 讨论 / Discussion

3.1 耦合反馈与不确定性压缩 / Coupling Feedback and Uncertainty Compression

The ~28% reduction in stress standard deviation from the decoupled (σ ≈ 49.2 MPa) to the coupled steady-state (σ ≈ 35.5 MPa) — a direct file-supported result of the forward UQ — is consistent with the type of negative feedback commonly described for compact coupled reactor systems: a plausible interpretation is that thermal-expansion-induced geometry changes modify the neutronic balance and partially offset the direct stress increase, although the relative contribution of each intermediate step cannot be isolated from the BNN's input–output map and has not been verified by a stepwise-decoupled numerical experiment. What the BNN-based forward UQ does establish is the net effect on the output uncertainty distribution — information that is inaccessible from single-physics or decoupled analyses alone.

从解耦（σ ≈ 49.2 MPa）到耦合稳态（σ ≈ 35.5 MPa）应力标准差降低约 28% ——这是前向 UQ 给出的直接可追溯结果——与紧凑耦合反应堆系统中常见的负反馈类型相一致：一种合理的解释是热膨胀引起的几何变化改变了中子学平衡、并部分抵消了直接应力增加，但各中间步骤的相对贡献不能仅凭 BNN 的输入–输出映射分离，也未通过逐步解耦数值实验独立验证。BNN 前向 UQ 所确立的是耦合对输出不确定性分布的净效应——这一信息在单物理或解耦分析中不可获取。

A practical implication is that performing uncertainty propagation at the decoupled level would overestimate tail stress risk. For safety-margin characterisation, this overestimate is conservative — it would not lead to unsafe decisions — but it would inflate the apparent safety penalty associated with material-property uncertainty, potentially leading to unnecessarily restrictive manufacturing tolerances.

实际含义是在解耦层次开展不确定性传播将高估尾部应力风险。对安全裕量表征，该高估是保守的——不会导致不安全决策——但会夸大与材料不确定性相关的安全惩罚，可能导致不必要的严格制造公差。

Chen et al. (2025, 核动力工程) observed qualitatively similar coupling-driven uncertainty effects in their HPR analysis; the present work extends this by providing quantitative characterisation through the BNN framework.

Chen et al.（2025, 核动力工程）在其 HPR 分析中观察到定性上类似的耦合驱动不确定性效应；本文通过 BNN 框架将其扩展为定量表征。

3.2 BNN 校准与模型选择 / BNN Calibration and Model Selection

The selection of the constraint-regularized BNN as the primary surrogate rests on two properties observed on the held-out test set. First, the monotonicity and inequality priors tighten the 90% predictive interval (mean MPIW reduced relative to the ELBO-only ablation reported in the Appendix) without degrading 90% coverage, indicating that the constraint terms sharpen predictive uncertainty rather than introducing systematic miscalibration. Second, the BNN provides a single posterior predictive distribution that propagates consistently through forward UQ, Sobol attribution, and MCMC posterior calibration, so that all downstream distribution-level quantities are computed against the same predictive object. This consistency property is the prerequisite for the analyses in Sections 2.2–2.4 and is not supplied by a point-estimate surrogate.

选择约束正则化 BNN 作为主代理，主要基于留出测试集上的两点观察。其一，单调性与不等式先验收紧 90% 预测区间（相对附录所列的纯 ELBO 消融，平均 MPIW 更窄），同时不损失 90% 覆盖率，说明约束项是在锐化预测不确定性而非引入系统性失校。其二，BNN 提供单一的后验预测分布，可在前向 UQ、Sobol 归因与 MCMC 后验标定之间一致传播，使所有下游分布级量均以同一预测对象为基础计算。该一致性属性是 §2.2–§2.4 各项分析的前置条件，点估计代理无法提供。

3.3 工程含义 / Engineering Implications

The Sobol pathway separation — stress dominated by $E_{\text{intercept}}$ ($S_1 \approx 0.56$) and k_{eff} dominated by α_{base} ($S_1 \approx 0.74$) — suggests two largely independent material-characterisation priorities for the coupled system. Sharpening the spread of $E_{\text{intercept}}$ on the input side would, on the BNN response surface, address the largest single first-order contribution to stress variability; sharpening the

spread of α_{base} would similarly address the largest contribution to k_{eff} variability. Because the two leading factors act on different outputs with limited cross-coupling at the first-order level, the analysis motivates treating stress-margin and reactivity-stability material characterisation as separable rather than as a single coupled optimisation. We frame these as model-driven indications of where material-level uncertainty reduction would be most informative for the coupled response, not as design directives.

Sobol 路径分离——应力由 $E_{intercept}$ 主导（$S_1 \approx 0.56$）、k_{eff} 由 α_{base} 主导（$S_1 \approx 0.74$）——提示耦合系统在材料层面存在两条基本独立的不确定性表征路径。在 BNN 响应面上，缩小 $E_{intercept}$ 的输入分布将作用于一阶应力变异中最大的单一贡献；同样地，缩小 α_{base} 的输入分布将作用于 k_{eff} 一阶变异中最大的贡献。由于两个主控因子分别作用于不同输出且在一阶层面交叉影响有限，本分析支持将应力裕量与反应性稳定性的材料表征视为可分而非须联合优化的两类任务。上述结论作为耦合响应中"在哪里减小材料级不确定性最具信息量"的模型驱动提示给出，而非设计指令。

The posterior calibration results further suggest that a sequential inference workflow is viable in principle: surrogate-based forward analysis under a prior characterises the parameter–response mapping at the population level, while subsequent observation-conditioned calibration contracts the inferred parameter distribution toward observation-compatible regions. This sequence aligns with the staged characterisation typical of advanced reactor programmes.

后验标定结果进一步提示序贯推断工作流在原理上可行：先验下的代理前向分析在分布层面刻画参数–响应映射，随后的观测条件标定将所推断的参数分布收敛至与观测相容的区域。该序贯结构与先进反应堆工程中典型的分阶段表征流程相吻合。

3.4 计算效率 / Computational Efficiency

The BNN produces one posterior-predictive draw (50 Monte Carlo weight samples) in 1.29×10^{-2} s on GPU, and a batched per-sample inference cost of 1.34×10^{-5} s (batch size 1024, throughput $\approx 7.5\times10^{4}$ samples/s). The coupled HF solver requires 2266 s ($\approx$ 38 min) per case, measured as the mean of three independent runs (std 1.34 s) on AMD EPYC 9654 (96-core) with 566 GB RAM and 2× RTX 5090. The resulting speedups are $\approx 1.76\times10^{5}$ for one posterior-predictive draw and $\approx$

1.69×10^{8} for batched per-sample inference. A 20,000-sample forward-UQ pass requires ≈ 0.3 s of GPU time (excluding I/O), against 4.5×10^{7} s of sequential HF cost; this is the operating cost on which the full probabilistic workflow is built.

BNN 代理模型每完成一次后验预测抽样（50 样本 MC，覆盖后验权重） GPU 时间为 1.29×10^{-2} s；批处理（1024 样本）单样本推断时间为 1.34×10^{-5} s（吞吐 ≈ 7.5×10^{4} 样本/s）。耦合高保真求解器单次案例求解耗时 2266 s（≈ 38 分钟，3 次独立运行均值，标准差 1.34 s；硬件：AMD EPYC 9654 96-Core，566 GB RAM，2× RTX 5090）。对应加速比为 ≈ 1.76×10^{5}（一次后验预测抽样相对 HF）与 ≈ 1.69×10^{8}（批处理单样本相对 HF）。20,000 样本前向 UQ 推断（不含 I/O）约需 0.3 s GPU 时间，对应顺序 HF 成本为 4.5×10^{7} s；该推断代价构成完整概率工作流的基础。

This speedup enables the full probabilistic workflow — 20,000-sample Monte Carlo propagation, 50-replication Sobol analysis, and 18-case MCMC calibration — to run within minutes of wall-clock time, making iterative coupled-model exploration computationally tractable for HPR systems with significant material uncertainty.

该加速使完整概率工作流——20,000 样本蒙特卡洛传播、50 次重复 Sobol 分析与 18 例 MCMC 标定——可在数分钟内完成，使具有显著材料不确定性的 HPR 系统的迭代耦合模型探索在计算上可行。

3.5 局限性 / Limitations

Several limitations scope the conclusions of this study.

以下局限界定了本研究结论的适用范围。

First, the BNN was trained on a fixed dataset of 2900 samples from a single HPR design point (MegaPower reference). Changes in geometry, fuel enrichment, or operating conditions would require retraining. The dataset generation cost (~2900 HF runs × ≈ 38 min ≈ ~76 sequential compute-days) is a one-time investment that is amortized across all subsequent UQ, sensitivity, and posterior analyses.

第一，BNN 在单一 HPR 设计点（MegaPower 参考设计）的 2900 个样本固定数据集上训练。几何、燃料富集度或运行工况变化需要重新训练。数据集生成成本（~2900 次 HF 运行 × ≈ 38 分钟 ≈ ~76 个顺序计算日）为一次性投资，可在后续所有 UQ、敏感性与后验分析中摊销。

Second, the MCMC calibration fixed 4 of the 8 parameters at their true values. Releasing additional parameters from the fixed set (for example the thermal-transport coefficient k_{slope}) would require independent informative observations — such as thermal conductivity measurements — that are not available in the current benchmark. The reported posterior contraction is therefore a lower bound on what full 8-dimensional calibration could achieve.

第二，MCMC 标定将 8 个参数中的 4 个固定在真实值。若进一步从固定集合中放开参数（例如热传导温度依赖系数 k_{slope}），需要额外的独立信息性观测（如热导率测量）作为似然约束，而当前基准并不具备该条件。因此所报告的后验收缩量是完整 8 维标定所能达到的下界。

Third, k_{eff} prediction on feature-tail OOD subsets is heterogeneous: on the α_{base} tail the k_{eff} R^2 drops to ≈ 0.29 (vs. an in-distribution global R^2 of 0.825 on the full test set), whereas stress retains $R^2 \geq 0.75$ across all four tested feature-tail splits. Extrapolation of k_{eff} beyond the training range of the dominant reactivity coefficient is therefore the weakest generalization axis of the surrogate. The anomalous pattern in which several OOD subsets report *higher* R^2 than their in-distribution counterparts arises from SST inflation in the tail subset (large target variance relative to MAE) rather than genuine improvement, and is discussed in Appendix A.

第三，k_{eff} 在特征尾部 OOD 子集上的预测表现不均匀：在 α_{base} 尾部，k_{eff} R^2 降至 ≈ 0.29（相对于整体测试集分布内 R^2=0.825），而应力在所测四个特征尾部子集上均保持 $R^2 \geq 0.75$。因此 k_{eff} 在主控反应性系数训练范围外的外推是代理泛化的最薄弱方向。部分 OOD 子集出现 R^2 *高于* 分布内对照的异常模式，源于尾部子集的 SST 膨胀（目标方差相对 MAE 偏大），并非真正的泛化改进；详见附录 A。

Fourth, the supplementary high-fidelity consistency check (Section 2.5) covers only the 18 stratified benchmark cases at posterior-mean parameter vectors and is not yet complete in the present revision; a larger-scale random-draw HF validation, decoupled from the calibration benchmark cases, would strengthen the surrogate-fidelity evidence and is left to future work.

第四，§2.5 的高保真一致性补充抽查仅覆盖 18 例分层基准在后验均值参数向量处的重新评估，且在本修订中尚未完成；与标定基准案例脱钩的更大规模随机抽样 HF 验证将进一步强化代理保真度的证据，留待后续工作。

4. 方法 / Methods

4.1 材料不确定性模型 / Material Uncertainty Model

The coupled HPR simulation involves eight material properties that enter the calculation through constitutive models for elasticity, thermal expansion, and heat conduction. These eight parameters were selected because they govern the three constitutive relationships that directly control the coupled thermo-mechanical-neutronic response; other material properties (e.g. fuel thermal conductivity, heat-pipe working-fluid parameters) are treated as fixed at their nominal values in the present study. The eight parameters are grouped by physical function into three categories.

HPR 耦合仿真涉及八个通过弹性、热膨胀和热传导本构模型进入计算的材料性质。选择这八个参数是因为它们控制着直接影响耦合热–力–中子响应的三组本构关系；其他材料性质（如燃料热导率、热管工质参数）在本研究中固定在标称值。按物理功能分为三组。

Structural mechanics parameters. The temperature-dependent Young's modulus of the SS316 monolith is modelled as a linear function of temperature: $E(T) = E_{slope} \cdot T + E_{intercept}$, where $E_{intercept}$ (nominal: 2×10^{11} Pa) determines the baseline stiffness and E_{slope} (nominal: -7×10^{7} Pa/K) captures its temperature softening. Poisson's ratio ν (nominal: 0.31) characterises the multi-axial stress state. These three parameters control stress through the generalised Hooke's law: $\sigma = C(E, \nu) \cdot \varepsilon$, where C is the fourth-order stiffness tensor.

结构力学参数。SS316 基体的温度相关弹性模量建模为温度的线性函数： $E(T) = E_{slope} \cdot T + E_{intercept}$，其中 $E_{intercept}$（标称值 2×10^{11} Pa）确定基础刚度，E_{slope}（标称值 -7×10^{7} Pa/K）

反映温度软化效应。泊松比 ν（标称值 0.31）表征多轴应力状态。三个参数通过广义胡克定律 $\sigma = C(E, \nu) \cdot \varepsilon$ 控制应力，C 为四阶刚度张量。

Thermal deformation parameters. The coefficient of thermal expansion is modelled as $\alpha(T) = \alpha_{base} + \alpha_{slope} \cdot T$, where α_{base} (nominal: 1×10^{-5} K^{-1}) is the expansion coefficient at the reference condition and α_{slope} (nominal: 5×10^{-9} K^{-2}) captures its temperature dependence. These parameters serve a dual role: they govern thermal strain $\varepsilon_th = \alpha \cdot \Delta T$, which is the loading source for thermal stress, and they determine the magnitude of core geometric expansion, which is the primary pathway through which material properties affect the neutronic balance.

热变形参数。热膨胀系数建模为 $\alpha(T) = \alpha_{base} + \alpha_{slope} \cdot T$，$\alpha_{base}$ （标称值 1×10^{-5} K^{-1}）为参考条件下膨胀系数，α_{slope}（标称值 5×10^{-9} K^{-2}）为温度依赖性。这两个参数具有双重角色：控制热应变 $\varepsilon_th = \alpha \cdot \Delta T$（应力的载荷源），同时决定堆芯几何膨胀的幅度——材料性质影响中子学平衡的主要路径。

SS316 thermal transport parameters. The thermal conductivity of the monolith is modelled as $k(T) = k_{ref} + k_{slope} \cdot (T - T_{ref})$, with three uncertain inputs: the reference conductivity k_{ref} (nominal: 23.2 W/(m·K)), the reference temperature T_{ref} (nominal: 923.15 K), and the temperature-dependence coefficient k_{slope} (nominal: 1/75 W/(m·K²)). Together these three parameters control the temperature gradient across the monolith, which is the physical intermediary between fission power deposition and thermal stress. For consistency with the code and data files released with this work, k_{ref} is named SS316_k_ref, T_{ref} is named SS316_T_ref, and k_{slope} is named SS316_alpha in the source; the physics symbols k_{ref}, T_{ref} and k_{slope} are used throughout the manuscript.

SS316 热传导参数。基体热导率建模为 $k(T) = k_{ref} + k_{slope} \cdot (T - T_{ref})$，对应三个不确定输入：参考热导率 k_{ref}（标称值 23.2 W/(m·K)）、参考温度 T_{ref}（标称值 923.15 K）、以及温度依赖系数 k_{slope}（标称值 1/75 W/(m·K²)）。三者共同决定基体内部温度梯度——裂变功率沉积与热应力之间的物理中间量。为与本文公开代码与数据保持一致，k_{ref}、T_{ref}、k_{slope} 在源代码中分别记作 SS316_k_ref、SS316_T_ref、SS316_alpha；正文统一使用物理符号 k_{ref}、T_{ref}、k_{slope}。

All eight parameters are assigned independent uniform priors with half-width 10% of the nominal design value: θ_k ~ U[0.9 θ_nom,k, 1.1 θ_nom,k] (Table 1). The ±10% range is a conventional uncertainty margin for manufactured stainless-steel properties and is consistent with the perturbation range used in Zhang et al. (2025, Energy). The same prior distribution is used throughout the pipeline: as the sampling distribution in forward UQ (Section 4.4), as the input domain for Sobol analysis (Section 4.5), and as the prior in posterior calibration (Section 4.6).

全部 8 个参数赋予独立均匀先验，半宽为标称值的 10%：θ_k ~ U[0.9 θ_nom,k, 1.1 θ_nom,k]（表 1）。±10% 区间是制造级不锈钢性质的常规不确定性裕量，与 Zhang et al.（2025, Energy）中使用的扰动范围一致。该先验分布在全流水线中统一使用：作为前向 UQ（4.4 节）的采样分布、Sobol 分析（4.5 节）的输入域和后验标定（4.6 节）的先验。

4.2 耦合高保真仿真 / Coupled High-Fidelity Simulation

The high-fidelity (HF) workflow couples OpenMC Monte Carlo neutronics with the FEniCS finite-element solver through a Picard fixed-point iteration scheme (Zhang et al. 2024). Starting from the eight material parameters and the reference core geometry, OpenMC computes the neutron multiplication factor k_{eff} and the spatially resolved fission power distribution. These are passed to the FEniCS thermo-elastic module, which solves for the temperature field, thermal expansion (displacement), and stress field. The updated geometry and temperature-dependent material cross-sections are then fed back to OpenMC for the next iteration. Convergence is declared when the maximum temperature difference between consecutive iterations falls below 1 K, with an under-relaxation factor of 0.1 applied to stabilise the iteration (Zhang et al. 2025, Energy).

高保真（HF）工作流通过 Picard 不动点迭代方案耦合 OpenMC 蒙特卡洛中子学与 FEniCS 有限元求解器（Zhang et al. 2024）。从八个材料参数和参考堆芯几何出发，OpenMC 计算中子增殖因子 k_{eff} 和空间分辨的裂变功率分布，传递给 FEniCS 热弹性模块求解温度场、热膨胀（位移）和应力场。更新的几何和温度相关材料截面反馈回 OpenMC 进行下一轮迭代。当连续迭代间最大温差低于 1 K 时宣告收敛，亚松弛因子 0.1 用于稳定迭代（Zhang et al. 2025, Energy）。

Two fidelity levels are retained from this iteration process. The decoupled single-pass prediction consists of the outputs after the first iteration (before coupling convergence): FEniCS receives the initial-geometry power distribution from OpenMC and produces thermal-structural outputs without feedback. The coupled steady-state prediction consists of the outputs after full convergence. Comparing these two levels isolates the effect of multi-physics coupling on the output distributions. The BNN is trained to predict both levels simultaneously.

从该迭代过程中保留两个精度层次。解耦单轮预测由第一轮迭代后的输出构成（耦合收敛前）：FEniCS 接收 OpenMC 的初始几何功率分布，产生无反馈的热力输出。耦合稳态预测由完全收敛后的输出构成。比较两个层次可分离多物理耦合对输出分布的影响。BNN 同时预测两个层次。

Each HF evaluation produces 15 scalar outputs. The 7 decoupled (iteration-1) outputs are: average fuel temperature, peak fuel temperature, peak monolith temperature, peak global stress, monolith mean temperature, core height after expansion, and heat-pipe wall displacement. The 8 coupled (iteration-2) outputs add k_{eff} to the same 7 thermal-structural quantities. The five primary outputs used in the posterior likelihood (Section 4.6) are: coupled peak global stress, coupled peak fuel temperature, coupled peak monolith temperature, coupled wall displacement, and k_{eff}.

每次 HF 评估产生 15 个标量输出。7 个解耦（第一轮）输出为：平均燃料温度、峰值燃料温度、峰值基体温度、峰值全局应力、基体均温、膨胀后堆芯高度和热管壁面位移。8 个耦合（第二轮）输出在相同 7 个热力量基础上增加 k_{eff}。后验似然（4.6 节）中使用的 5 个主输出为：耦合峰值全局应力、耦合峰值燃料温度、耦合峰值基体温度、耦合壁面位移和 k_{eff}。

A dataset of n = 2900 samples was generated by Latin hypercube sampling over the 8-dimensional prior. A small number of samples produced non-converged or NaN outputs in the coupled iteration; these were excluded prior to surrogate training. The remaining samples were split into training (2029, ~70%), validation (436, ~15%), and test (435, ~15%) sets with a fixed random seed (seed = 2026), ensuring reproducibility across all experiments. The BNN does not resolve intermediate coupling quantities (e.g. spatially resolved power or temperature fields); it maps directly from the 8 input parameters to the 15 scalar outputs.

通过拉丁超立方采样在 8 维先验空间上生成 n = 2900 个样本。少量样本在耦合迭代中产生未收敛或 NaN 输出，在代理训练前予以排除。剩余样本以固定随机种子（seed = 2026）划分为训练（2029，~70%）、验证（436，~15%）和测试（435，~15%）集，确保所有实验的可重复性。BNN 不解析中间耦合量（如空间分辨的功率或温度场），直接从 8 个输入参数映射到 15 个标量输出。

4.3 贝叶斯神经网络代理 / Bayesian Neural Network Surrogate

The surrogate is a Bayesian neural network (BNN). Unlike a deterministic network that returns a single point prediction, a BNN places a prior distribution over network weights and, after training, represents knowledge of the input–output map through a posterior distribution over weights. The resulting posterior predictive distribution serves as the common probabilistic object that all downstream analyses (forward UQ, sensitivity attribution, MCMC calibration) operate on. The same distribution can additionally be decomposed into a weight-uncertainty (epistemic) component and an observation-noise (aleatoric) component; this decomposition is reported in the Appendix as a diagnostic of the predictive variance, but is not used to drive the main-text analyses.

代理模型为贝叶斯神经网络（BNN）。与返回单点预测的确定性网络不同，BNN 对网络权重引入先验分布，并在训练后以权重的后验分布表示输入–输出映射的知识。由此得到的后验预测分布作为下游各类分析（前向 UQ、敏感性归因、 MCMC 标定）共用的概率对象。该分布另可分解为权重不确定性（认识论）分量与观测噪声（偶然）分量；该分解在附录中作为预测方差的诊断信息报告，不参与正文主线分析的驱动。

Architecture. The BNN uses a fully connected backbone (depth 3–8 layers, width 64–256 per layer, SiLU activation) with two output heads at the final layer: a mean head producing $\mu \in \mathbb{R}^{15}$ and a log-variance head producing $\log \sigma^2_obs \in \mathbb{R}^{15}$ (clamped to [−20, 5] for numerical stability) that models aleatoric observation noise. The 8 input parameters and 15 output targets are standardised to zero mean and unit variance using training-set statistics before training.

架构。BNN 采用全连接主干（深度 3–8 层，每层宽度 64–256，SiLU 激活），末端并列两个输出头：均值头输出 $\mu \in \mathbb{R}^{15}$，对数方差头输出 $\log \sigma^2_{obs} \in \mathbb{R}^{15}$（限幅至 $[-20, 5]$），后者用于建模偶然观测噪声。8 个输入参数和 15 个输出目标在训练前使用训练集统计量标准化为零均值单位方差。

Probabilistic formulation. Let f_w denote the network with weights w. The model assumes a Gaussian likelihood per output channel:

$$y_j \mid \theta, w \sim N(\mu_{j,w}(\theta), \sigma^2_{obs,j,w}(\theta)), \quad j = 1, \ldots, 15,$$

with an isotropic Gaussian weight prior $p(w) = N(0, \sigma^2_p I)$, where σ_p is treated as a hyperparameter (default 1.0; Optuna-searchable). Training then produces an approximate posterior $q(w) \approx p(w \mid D)$ over the weights, where D denotes the training set. The posterior predictive distribution for a new input θ is obtained by marginalising over $q(w)$:

$$p(y \mid \theta, D) = \int p(y \mid \theta, w)\, q(w)\, dw \approx (1/M) \Sigma_{m=1..M} p(y \mid \theta, w^{(m)}), \quad w^{(m)} \sim q(w).$$

概率形式。记网络为 f_w（权重 w），模型对每个输出通道假设高斯似然：

$$y_j \mid \theta, w \sim N(\mu_{j,w}(\theta), \sigma^2_{obs,j,w}(\theta))，j = 1, \ldots, 15，$$

权重先验为各向同性高斯 $p(w) = N(0, \sigma^2_p I)$，$\sigma_p$ 作为超参（默认 1.0，可经 Optuna 搜索）。训练得到权重的近似后验 $q(w) \approx p(w \mid D)$，D 为训练集。新输入 θ 处的后验预测分布通过对 $q(w)$ 边缘化获得：

$$p(y \mid \theta, D) = \int p(y \mid \theta, w)\, q(w)\, dw \approx (1/M) \Sigma_{m=1..M} p(y \mid \theta, w^{(m)})，\quad w^{(m)} \sim q(w)。$$

Inference scheme. The BNN is trained by mean-field variational inference (MFVI) in the Bayes-by-Backprop formulation (Blundell et al. 2015 [10]). Each weight is endowed with an independent Gaussian variational posterior, $q(w) = \Pi_\ell N(\mu_\ell, \mathrm{softplus}(\rho_\ell)^2)$, and the variational parameters $\{\mu_\ell, \rho_\ell\}$ are optimised by maximising the ELBO with the reparameterisation gradient. The training objective is the negative evidence lower bound (ELBO):

$$L(\varphi) = - E_{q_\varphi(w)}[\log p(D \mid w)] + KL[q_\varphi(w) \parallel p(w)],$$

where φ collects the variational parameters. The first term is a heteroscedastic Gaussian negative log-likelihood over the training set; the second term regularises $q(w)$ toward the prior. A KL annealing schedule is used so that the data term dominates early in training.

推断方案。BNN 采用 Bayes-by-Backprop 形式的 mean-field 变分推断（MFVI，Blundell et al. 2015 [10]）。每个权重配备独立的高斯变分后验 $q(w) = \Pi_\ell N(\mu_\ell, \mathrm{softplus}(\rho_\ell)^2)$，变分参数 $\{\mu_\ell, \rho_\ell\}$ 通过最大化 ELBO 并使用重参数化梯度优化。训练目标为负证据下界（ELBO）：

$$L(\varphi) = -E_{q_\varphi(w)}[\log p(D \mid w)] + KL[q_\varphi(w) \parallel p(w)],$$

φ 为变分参数。第一项为训练集上的异方差高斯负对数似然，第二项把 $q(w)$ 向先验正则化。采用 KL 退火使训练前期由数据项主导。

Predictive mean and variance. At inference, $M = 50$ weight samples $w^{(m)} \sim q(w)$ are drawn (Sobol analysis uses $M = 30$ and the MCMC likelihood uses $M = 20$ for computational efficiency, as both already carry their own Monte Carlo averaging). The posterior predictive mean and variance for output j at input θ are:

$$\hat{\mu}_j(\theta) = (1/M) \Sigma_m \mu_{j, w^{(m)}}(\theta),$$

$$\hat{\mathrm{Var}}_j(\theta) = (1/M) \Sigma_m \sigma^2_{obs,j,w^{(m)}}(\theta) \quad \text{[observation-noise]} + (1/M) \Sigma_m (\mu_{j,w^{(m)}}(\theta) - \hat{\mu}_j(\theta))^2 \quad \text{[weight-uncertainty]}.$$

The 90% predictive interval is constructed from the empirical 5th and 95th percentiles of the M predictive draws per output. The two terms in $\hat{\mathrm{Var}}_j(\theta)$ provide an aleatoric/epistemic decomposition of the predictive variance, which is reported as a diagnostic in the Appendix; the main-text analyses use the full posterior predictive distribution rather than either component in isolation.

预测均值与方差。推断时抽取 $M = 50$ 个权重样本 $w^{(m)} \sim q(w)$（Sobol 分析使用 $M = 30$、MCMC 似然使用 $M = 20$ 以兼顾效率——两者本身已含大量蒙特卡洛平均）；输入 θ 处输出 j 的后验预测均值与方差为：

$$\hat{\mu}_j(\theta) = (1/M) \Sigma_m \mu_{j, w^{(m)}}(\theta),$$

$\hat{Var}_j(\theta) = (1/M) \Sigma_m \sigma^2_{obs,j,w^{(m)}}(\theta)$ [观测噪声] + $(1/M) \Sigma_m (\mu_{j,w^{(m)}}(\theta) - \hat{\mu}_j(\theta))^2$ [权重不确定性]。

90% 预测区间由每个输出的 M 个预测样本的第 5–95 百分位得到。$\hat{Var}_j(\theta)$ 中的两项提供预测方差的偶然/认识论分解，作为附录中报告的诊断信息；正文主线分析使用完整的后验预测分布，而非其中任一单独分量。

Hyperparameters were independently optimised via Optuna using a TPE sampler with median pruning. The search space included: depth (3–8 layers), width (64–256), learning rate (10^{-4}–3×10^{-3}), batch size ({32, 64, 128}), gradient clipping (0.5–5.0), the prior scale σ_p, the initial variational posterior scale, and the KL annealing schedule. Training ran for up to 300 epochs with early stopping (patience 25–30, minimum improvement 10^{-6}) on validation ELBO. The final checkpoint was selected as the trial with the lowest validation NLL (posterior predictive). All models used Adam optimisation and Xavier uniform initialisation of the variational mean parameters.

超参数通过 Optuna 使用 TPE 采样器和中位数剪枝独立优化。搜索空间包括：深度（3–8 层）、宽度（64–256）、学习率（10^{-4}–3×10^{-3}）、批大小（{32, 64, 128}）、梯度裁剪（0.5–5.0）、先验尺度 σ_p、变分后验初始尺度以及 KL 退火方案。训练至多 300 个 epoch，基于验证集 ELBO 早停（耐心 25–30，最小改进 10^{-6}）。最终检查点选取验证集后验预测 NLL 最低的试验。所有模型使用 Adam 优化器与 Xavier 均匀初始化变分均值参数。

4.4 前向不确定性传播方法 / Forward UQ Methodology

Forward uncertainty propagation quantifies how prior parameter uncertainty maps to output uncertainty through the BNN. 20,000 samples are drawn from the joint prior (Section 4.1) by Latin hypercube sampling. For each sample θ_i, the BNN returns the posterior predictive mean $\hat{\mu}(\theta_i)$ for all 15 outputs (marginalising over M weight samples). The empirical distribution of $\{\hat{\mu}(\theta_i)\}_{i=1}^{20000}$ constitutes the prior-predictive uncertainty propagation.

前向不确定性传播量化先验参数不确定性如何通过 BNN 映射到输出不确定性。从联合先验（4.1 节）通过拉丁超立方采样抽取 20,000 个样本。对每个 θ_i， BNN 返回 15 个输出

的后验预测均值 $\hat{\mu}(\theta_i)$（对 M 个权重样本边缘化）。 $\{\hat{\mu}(\theta_i)\}$ 的经验分布构成先验预测不确定性传播。

This protocol ($y = \hat{\mu}(\theta) = E_w[\mu_w(\theta)]$) deliberately isolates parametric uncertainty from the BNN's epistemic+aleatoric predictive spread. Because the BNN's posterior predictive variance already models both the observation-noise and weight-uncertainty contributions, including it in the forward propagation ($y = \hat{\mu}(\theta) + \hat{\sigma}(\theta) \cdot \varepsilon$, $\varepsilon \sim N(0,1)$) would mix the parametric prior with the surrogate's own uncertainty. The main-text results report the $\hat{\mu}$-only protocol so that all variance in the output distribution is attributable to input-parameter variation. The predictive variant is reported in the appendix as a consistency check; it yields broader distributions but preserves the same qualitative conclusions.

该协议（$y = \hat{\mu}(\theta) = E_w[\mu_w(\theta)]$）有意隔离参数不确定性与 BNN 自身的认识论+偶然预测展宽。由于 BNN 后验预测方差已同时建模观测噪声与权重不确定性，若在前向传播中包含它（$y = \hat{\mu}(\theta) + \hat{\sigma}(\theta) \cdot \varepsilon$）会将参数先验与代理自身不确定性混合。正文结果报告 $\hat{\mu}$-only 协议，使输出分布中的所有方差可归因于输入参数变异。预测变体在附录中作为一致性检验报告；它产生更宽的分布但保持相同的定性结论。

Summary statistics of $\{\hat{\mu}(\theta_i)\}$ (mean, standard deviation, percentiles) are reported in Section 2.2 for the primary outputs. Distribution-level characterisation — i.e., how multiphysics coupling reshapes the output distribution relative to the decoupled prediction — is the role of forward UQ in this work; threshold-based exceedance statistics are not part of the main-text analysis.

$\{\hat{\mu}(\theta_i)\}$ 的汇总统计量（均值、标准差、分位数）在 §2.2 中报告。本文前向 UQ 的角色为分布级表征——即多物理耦合相对解耦预测如何重塑输出分布；正文不展开基于阈值的超阈统计。

4.5 Sobol 敏感性分析方法 / Sobol Sensitivity Methodology

Variance-based global sensitivity analysis is built on the Sobol decomposition of a square-integrable output $Y = \hat{\mu}(\theta)$ of independent inputs $\theta = (\theta_1, \ldots, \theta_d)$:

$$\mathrm{Var}(Y) = \Sigma_i V_i + \Sigma_{i<j} V_{ij} + \ldots + V_{1\ldots d},$$

where $V_i = Var_{\theta_i}(E[Y \mid \theta_i])$ is the main-effect variance of input i and the higher-order terms collect pairwise and multi-way interactions. The first-order and total-order Sobol indices are

$$S_i = V_i / Var(Y), \quad S_{T,i} = 1 - Var_{\theta_{\sim i}}(E[Y \mid \theta_{\sim i}]) / Var(Y).$$

S_i measures the fraction of output variance attributable to input i alone; $S_{T,i}$ collects all effects involving input i, including interactions. A large gap between $S_{T,i}$ and S_i indicates substantial interaction effects; when $S_{T,i} \approx S_i$, input i acts primarily through an independent channel.

基于方差的全局敏感性分析建立在 Sobol 分解之上。对于独立输入 $\theta = (\theta_1, \ldots, \theta_d)$ 的平方可积输出 $Y = \hat{\mu}(\theta)$，有：

$$Var(Y) = \Sigma_i V_i + \Sigma_{i<j} V_{ij} + \ldots + V_{1\ldots d},$$

其中 $V_i = Var_{\theta_i}(E[Y \mid \theta_i])$ 为输入 i 的主效应方差，高阶项收集两两和多项交互。一阶与总阶 Sobol 指数分别为：

$$S_i = V_i / Var(Y), \quad S_{T,i} = 1 - Var_{\theta_{\sim i}}(E[Y \mid \theta_{\sim i}]) / Var(Y)。$$

S_i 度量仅由输入 i 单独引起的输出方差比例；$S_{T,i}$ 汇集所有涉及输入 i 的效应（含交互）。$S_{T,i}$ 与 S_i 之间的较大差距表明存在显著的交互效应；当 $S_{T,i} \approx S_i$ 时，输入 i 主要通过独立通道起作用。

Indices are estimated using the Saltelli quasi-random sampling scheme with the Jansen estimator. Two independent $N_S \times 8$ matrices A and B are drawn from the prior ($N_S = 4096$ per matrix). For each input dimension i, a cross-matrix $A_B^{(i)}$ is formed by replacing the i-th column of A with that of B. The total number of BNN posterior-predictive-mean evaluations per replication is $N_S \times (8 + 2) = 40{,}960$. The BNN posterior predictive mean $\hat{\mu}(\theta)$ serves as the deterministic evaluator, for the same reason as in forward UQ: isolating parametric sensitivity from surrogate uncertainty.

采用 Saltelli 准随机采样方案结合 Jansen 估计量估计指数。从先验中抽取两个独立的 $N_S \times 8$ 矩阵 A 和 B（每矩阵 $N_S = 4096$）。对每个输入维度 i，将 A 的第 i 列替换为 B 的第 i

列形成交叉矩阵 A_B^(i)。每次重复的 BNN 后验预测均值评估总数为 N_S × (8 + 2) = 40,960。BNN 后验预测均值 $\hat{\mu}(\theta)$ 充当确定性评估器，原因与前向 UQ 相同：隔离参数敏感性与代理不确定性。

To quantify sampling uncertainty in the estimated indices, the analysis is independently repeated 50 times with different random seeds. The 90% confidence interval for each index is computed as $\bar{S}_i \pm 1.645 \times SE$, where SE is the standard error across the 50 replications. A factor is considered a stable contributor only if the lower bound of its 90% CI exceeds zero; factors whose CI includes zero are not interpreted as meaningful contributors regardless of their point estimate.

为量化估计指数中的采样不确定性，分析以不同随机种子独立重复 50 次。每个指数的 90% 置信区间计算为 $\bar{S}_i \pm 1.645 \times SE$，SE 为 50 次重复的标准误差。仅 90% CI 下界超过零的因子被视为稳定贡献者；CI 包含零的因子无论点估计值如何均不被解释为有意义的贡献者。

4.6 后验标定方法 / Posterior Calibration Methodology

Posterior calibration uses Bayesian inference to update the prior parameter distribution given observed coupled outputs, quantifying how much parameter uncertainty can be reduced by measurement-like information. Given a prior $\pi(\theta)$ and a likelihood $p(y_{obs} \mid \theta)$ induced by the BNN posterior predictive mean and the observation-noise model, the posterior over calibrated parameters is

$$p(\theta \mid y_{obs}) \propto p(y_{obs} \mid \theta) \cdot \pi(\theta).$$

The posterior is represented by MCMC samples $\{\theta^{(m)}\}_{m=1}^{M}$ and is used downstream as an updated parameter distribution that can be re-evaluated through the BNN posterior predictive function for any output of interest.

后验标定使用贝叶斯推断在给定观测耦合输出的条件下更新先验参数分布，量化类测量信息可以减少多少参数不确定性。给定先验 $\pi(\theta)$ 以及由 BNN 后验预测均值与观测噪声模型共同给出的似然 $p(y_{obs} \mid \theta)$，标定参数的后验为：

$$p(\theta \mid y_{obs}) \propto p(y_{obs} \mid \theta) \cdot \pi(\theta)。$$

后验通过 MCMC 样本 $\{\theta^{(m)}\}_{m=1}^{M}$ 表示，下游作为更新后的参数分布使用，并可针对任意感兴趣输出经 BNN 后验预测函数重新评估。

Four parameters are selected for calibration: $E_{intercept}$, α_{base}, α_{slope}, and k_{ref}. This subset was chosen based on the Sobol ranking: these four parameters have the largest S_1 values for stress and/or k_{eff}, collectively accounting for the majority of the explained variance. The remaining four parameters (E_{slope}, T_{ref}, k_{slope}, ν) are fixed at their true test-case values. Fixing these parameters is a simplification that assumes they could be independently measured; relaxing this assumption to a full 8-dimensional calibration would require additional independent observations to constrain the expanded parameter space.

选择四个参数进行标定：$E_{intercept}$、α_{base}、α_{slope} 和 k_{ref}。该子集基于 Sobol 排序选择：这四个参数对应力和/或 k_{eff} 具有最大的 S_1 值，共同解释了大部分方差。其余四个参数（E_{slope}、T_{ref}、k_{slope}、ν）固定在真实测试案例值。固定这些参数是一种简化假设，假定它们可被独立测量；将该假设放松为完整 8 维标定需要额外的独立观测来约束扩展的参数空间。

Benchmark cases are drawn from the held-out test split to ensure that the BNN has not seen the true parameter–output pairs during training. A total of 18 benchmark cases are selected by stratified sampling across three stress bands: six low-stress, six intermediate-stress, and six high-stress cases. The stratification is intended to cover the operating range of the surrogate rather than to evaluate any threshold-based acceptance criterion.

基准案例从留出测试集中抽取，确保 BNN 在训练中未见过真实参数–输出对。通过在三个应力带中分层采样共选取 18 个基准案例：低应力、中应力、高应力各 6 例。该分层旨在覆盖代理的运行范围，而非用于评估任何基于阈值的接受准则。

For each benchmark case, the "observations" are the true HF outputs of 5 primary coupled quantities (peak global stress, peak fuel temperature, peak monolith temperature, wall displacement, and k_{eff}) corrupted by 2% multiplicative Gaussian noise: $y_obs = y_true + \varepsilon$, $\varepsilon \sim N(0, (0.02 \times |y_true|)^2)$. This noise level simulates realistic measurement uncertainty.

对每个基准案例，"观测"为 5 个主耦合输出（峰值全局应力、峰值燃料温度、峰值基体温度、壁面位移和 k_{eff}）的真实 HF 输出加 2% 乘性高斯噪声： $y_obs = y_true + \varepsilon$，$\varepsilon \sim N(0, (0.02 \times |y_true|)^2)$。该噪声水平模拟实际测量不确定性。

The posterior is sampled using Metropolis-Hastings MCMC with 4 independent chains per case, each initialised from a different random draw from the prior to enable a Gelman-Rubin convergence diagnostic. The proposal distribution is a Gaussian centred at the current state with scale $0.40 \times \sigma_prior$ for each parameter, where σ_prior is the prior standard deviation. Each chain runs for 8000 iterations; the first 2000 are discarded as burn-in and every 5th sample is retained, yielding 1200 posterior samples per chain and 4800 per case after pooling. The log-likelihood for a candidate parameter vector θ^* uses the BNN posterior predictive mean $\hat{\mu}_j(\theta^*)$ (marginalised over weight samples) as the forward-model surrogate:

后验通过 Metropolis-Hastings MCMC 采样，每个案例独立运行 4 条链，各链从先验中随机抽取不同起点初始化，以便计算 Gelman-Rubin 收敛诊断。提议分布为以当前状态为中心的高斯分布，每个参数的尺度为 $0.40 \times \sigma_prior$ （σ_prior 为先验标准差）。每条链运行 8000 步迭代；前 2000 步作为烧入期丢弃，每 5 步保留 1 个样本，每条链 1200 个后验样本，合并后每个案例共 4800 个。候选参数向量 θ^* 的对数似然使用对权重样本边缘化后的 BNN 后验预测均值 $\hat{\mu}_j(\theta^*)$ 作为前向模型代理：

$$\log L(y_obs \mid \theta^*) = -(1/2)\, \Sigma_j\, [(y_obs,j - \hat{\mu}_j(\theta^*))^2 / \sigma^2_obs,j + \log \sigma^2_obs,j]$$

where $\sigma_obs,j = 0.02 \times |y_true,j|$ is the observation noise standard deviation. Acceptance rates across the 18 benchmark cases are reported in Section 2.4. Convergence is assessed by the Gelman-Rubin statistic $\hat{R}$ across the 4 chains per case, supplemented by trace-plot inspection (Fig. A1). Cases are accepted as converged when $\hat{R} < 1.1$ for every calibrated parameter; the detailed per-case $\hat{R}$ values are reported in Appendix F.

其中 $\sigma_obs,j = 0.02 \times |y_true,j|$ 为观测噪声标准差。18 个基准案例的接受率在第 2.4 节报告。收敛以每案例 4 链的 Gelman-Rubin 统计量 $\hat{R}$ 为主评估，trace plot 为辅（图 A1）；当所有标定参数 $\hat{R} < 1.1$ 时该案例判为收敛，逐案例 $\hat{R}$ 见附录 F。

Posterior summaries reported in Section 2.4 (parameter-marginal 90% coverage and the posterior-mean BNN stress vs. true stress) are computed entirely on the surrogate side from $\{\theta^{(m)}\}_{m=1}^{M}$. The supplementary high-fidelity consistency check described in Section 2.5 re-evaluates the coupled solver at posterior-mean parameter vectors as an independent verification; it is not part of the calibration pipeline and does not constitute a systematic validation of surrogate generalisation.

§2.4 报告的后验汇总量（参数边际 90% 覆盖率与后验均值处的 BNN 应力对真实应力）完全在代理侧由 $\{\theta^{(m)}\}_{m=1}^{M}$ 计算。§2.5 所述的高保真一致性补充抽查在后验均值参数向量处重新评估耦合求解器，作为独立核验；该抽查不属于标定流水线，亦不构成对代理泛化能力的系统性验证。

5. 结论 / Conclusion

This work presented a Bayesian neural network probabilistic surrogate for coupled neutronics–thermal–structural analysis in a heat-pipe-cooled reactor, together with a forward-UQ / Sobol / MCMC workflow that operates on a single posterior predictive distribution. The principal findings are:

本文提出了面向热管冷却反应堆耦合中子–热–力分析的贝叶斯神经网络概率代理，并在同一后验预测分布之上组织前向 UQ、Sobol 与 MCMC 工作流。主要发现如下：

(1) The BNN attained stress $R^2 = 0.910$ with conservatively over-covered 90% predictive intervals ($PICP_{90} = 0.998$) on the held-out test set, providing a single posterior predictive distribution that is then used without retraining or recalibration across forward UQ, sensitivity and posterior calibration.

（1）BNN 在留出测试集上达到应力 $R^2 = 0.910$，90% 后验预测区间呈保守偏覆盖（$PICP_{90} = 0.998$），提供单一后验预测分布，并在不重训、不重校准的前提下被前向 UQ、敏感性与后验标定共同复用。

(2) Multiphysics coupling reshapes the material-parameter-driven stress distribution: the standard deviation contracts by ≈ 28% (decoupled $\sigma \approx 49.2$ MPa → coupled $\sigma \approx 35.5$ MPa) and the median

moves to lower values relative to the decoupled prediction, consistent with negative neutronics–thermal feedback that is invisible to decoupled analysis.

（2）多物理耦合重塑了材料参数驱动的应力分布：标准差收缩 ≈ 28% （解耦 $\sigma \approx 49.2$ MPa → 耦合 $\sigma \approx 35.5$ MPa），中位数相对解耦预测向低值移动，与解耦分析中不可见的中子–热负反馈一致。

(3) Stress and k_{eff} are governed by largely separated dominant pathways at the first-order level: $E_{intercept}$ ($S_1 \approx 0.56$, ~59% of first-order stress variance) for stress and α_{base} ($S_1 \approx 0.74$) for k_{eff}, with limited cross-interaction. This suggests that stress-margin and reactivity-stability material characterisation can be treated as two largely independent priorities.

（3）在一阶层面，应力与 k_{eff} 由基本独立的主控路径支配：应力为 $E_{intercept}$（$S_1 \approx 0.56$，约占一阶应力方差 59%），k_{eff} 为 α_{base} （$S_1 \approx 0.74$），两者交叉作用有限。该结果提示应力裕量与反应性稳定性的材料表征可视为两条基本独立的工作。

(4) MCMC calibration against noisy observations contracts the inferred parameter distribution toward observation-compatible regions of parameter space, with a 90%-CI coverage of 0.889 across 18 stratified benchmark cases and a posterior-mean BNN stress that tracks the true stress with MAE = 6.6 MPa (bias +2.5 MPa, stress range 133–194 MPa). A supplementary high-fidelity consistency check at the posterior-mean parameter vectors is described in Section 2.5; it is in progress and will be reported in the next revision.

（4）MCMC 在含噪观测下的标定将所推断的参数分布收敛至与观测相容的参数空间区域：18 例分层基准案例的 90%-CI 覆盖率为 0.889，后验均值处的 BNN 应力与真实应力吻合（MAE = 6.6 MPa，偏差 +2.5 MPa，应力范围 133–194 MPa）。在后验均值参数向量处的高保真一致性补充抽查见 §2.5，当前正在进行中，将在下一次修订中报告。

The $\approx 1.76 \times 10^5$ speedup per posterior-predictive draw ($\approx 1.69 \times 10^8$ batched per-sample) relative to a 2266 s coupled HF case on AMD EPYC 9654 makes the end-to-end probabilistic workflow tractable at the surrogate level, providing a unified BNN posterior predictive layer between expensive coupled multiphysics simulation and downstream probabilistic analysis in coupled HPR systems.

相对 AMD EPYC 9654 上 2266 s 的耦合 HF 单案例，每次后验预测抽样加速约 1.76×10^5 倍（批处理单样本约 1.69×10^8），使端到端概率工作流在代理层级上可行，为耦合 HPR 系统中昂贵的耦合多物理仿真与下游概率分析之间提供一个统一的 BNN 后验预测层。

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附录 / Appendices（内容摘要）

A. OOD 评估（四特征尾部：应力 OOD $R^2 \approx 0.75$–0.79 稳定；k_{eff} OOD R^2 跨特征为 0.29–0.80，α_{base} 尾部最低；附 SST 膨胀说明） B. 无约束 BNN 消融（去除单调性与不等式先验，仅保留 ELBO；与主代理的 R^2 / $PICP_{90}$ / $MPIW_{90}$ / 秩单调性对比） C. 完整 Sobol 结果（8 输入 × 2 输出，含 S_T 指数与 CI） D. 全部 15 输出量精度表（FIG A5: 分输出精度图） E. MCMC 迹图（FIG A1）+ 多链 $\hat{R}$ 诊断 F. 先验–后验边缘分布（FIG A2） G. BNN 校准可靠性曲线（FIG A3） H. 预测方差分解诊断（认识论 vs 偶然分量在测试集上的相对幅度，

FIG A4） I. 先验–后验预测分布对比（FIG A6） J. BNN 架构示意图（FIG A7） K. 训练曲线（FIG A8）——BNN ELBO 演化与无约束消融对比 L. 速度基准详情表（TABLE L1） M. 前向 UQ predictive variant 结果对比（$\hat{\mu}$-only vs full posterior predictive） N. BNN 推断方案消融（MFVI vs MC-Dropout vs Ensemble，若启用）